

When Should an Alert Threshold Learn?

Admission-separated recalibration as a decision-time action for time-series anomaly alerting

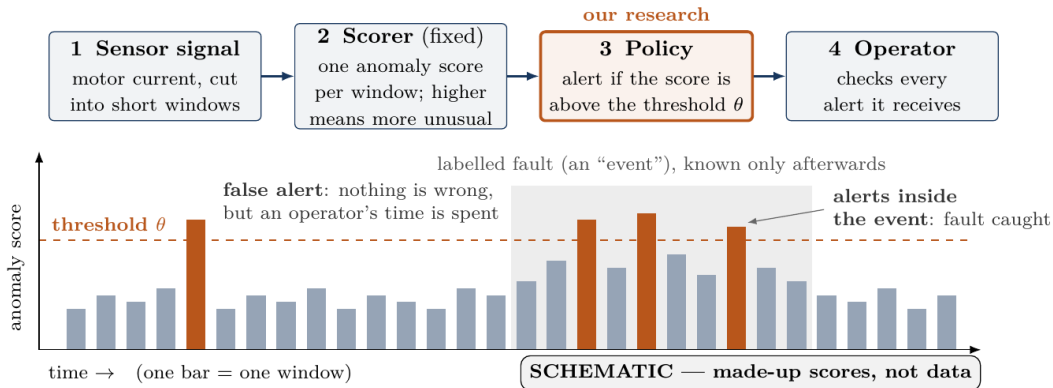
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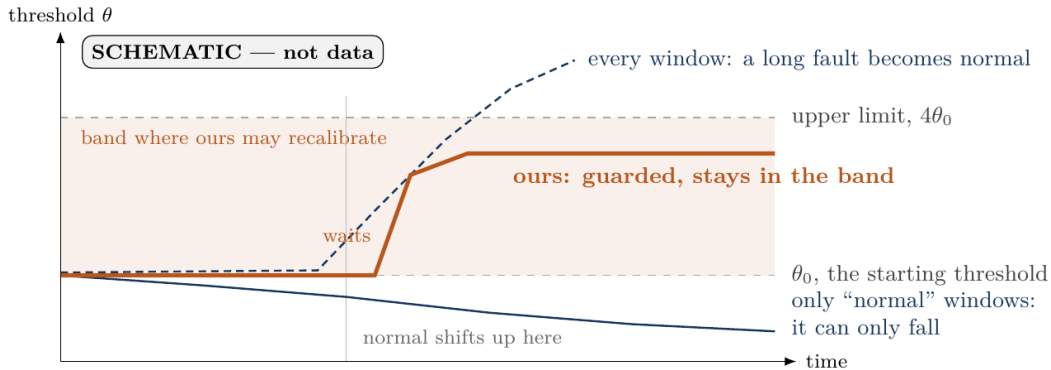
How alerting works: a score per window, a threshold, an operator

Our data: SKAB, a public benchmark recorded on a laboratory water-circulation rig. We use one sensor, the current drawn by its electric motor. **Two ways to fail:** miss a real fault, or raise false alerts. Each false alert costs an operator's time; too many, and people stop trusting the alerts.



Normal drifts, so thresholds learn, but from which windows?

When normal **shifts up** (a repair, a new load), a fixed threshold floods the operator with false alerts, so systems **recalibrate**: they re-set the threshold from recent scores. **Which windows should it learn from?** The two obvious answers both fail.



A threshold that learns only from “normal” windows can only fail

A threshold keeps its last 3 admitted scores, sets itself to their maximum, and admits a new score **only if it judged that score normal**. Score 9 is normal at step 1 and an alert at step 6: the alerts come from the threshold, not the signal.

Step	Score	Judged against	Decision	Threshold after
1	9	10	normal, admitted	9
2	12	9	alert, rejected	9
3	8	9	normal, admitted	9
4	7	9	normal, admitted	9
5	6	9	normal, admitted	8
6	9	8	alert , rejected	8

Our direction: recalibration as a guarded, auditable action

Recalibrate only as a deliberate decision, from data chosen **neither by the threshold itself nor unconditionally**, and judge every policy on **identical scores**. Tags: PROVED = theorem · MEASURED = saved, recomputed result · IMPLEMENTED = tested code · PROPOSED = not yet tested.

Evidence

What we claim

PROVED

C1: learning only from “normal” windows means the threshold never rises

MEASURED

C2: on SKAB that costs $16.2\times$ and $3.9\times$ the alert windows

MEASURED

C3: the scorer, not the policy, limits detection

IMPLEMENTED

C4: a four-action controller, recalibration bounded to $[\theta_0, 4\theta_0]$

MEASURED

C5: identical to fixed on SKAB, so no harm and no benefit

MEASURED

C6: under stress it works on a step and fails on a ramp

PROPOSED

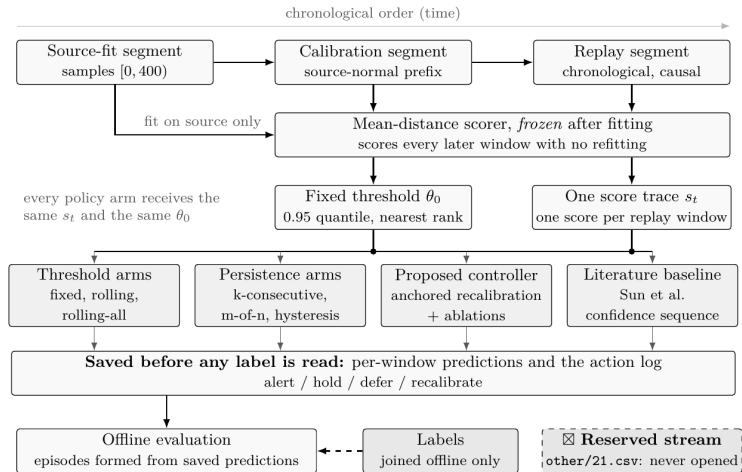
C7: a benefit on real regime changes, decided by a kill test

Isn't this just X? Each ingredient exists; the combination does not

Open in the nine papers checked: action policies compared on one frozen score trace. A bounded search, not proof of absence.

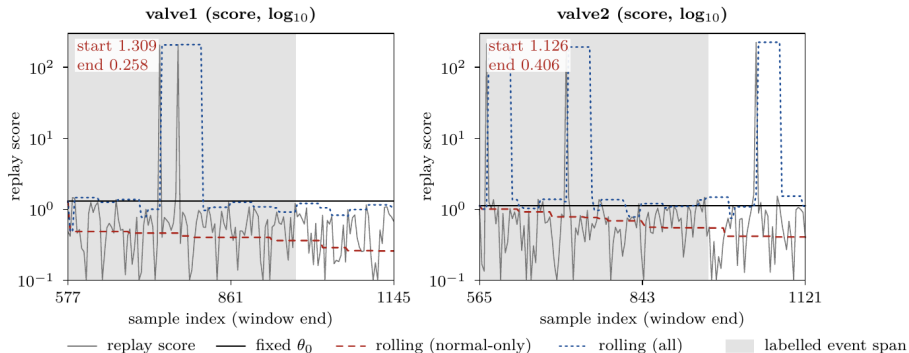
Closest work	Already does	Ours differs
FITNESS (ICML 2022)	Shows dropping self-flagged anomalies fails	We claim only the ratchet proof
Sun et al. (ICML 2024)	Abstains with guarantees	Re-implemented and measured
Perini & Davis	Rejects on a fixed stability threshold	Defer is charged to coverage
DDADE, ADAPTS, CDDIA	Retrain the model on drift	Scorer frozen; only actions change
Hu (in press)	Evaluates alerts by workload	Same frame, plus a controller
SEAD, Dynamic-XY	New score; suppression from alert history	One score trace, explicit actions

Every policy sees identical scores; labels arrive only afterwards



Proved, then observed: the self-referential threshold only falls

PROVED **Proposition 1.** Admit s only if $s \leq \theta$ and set θ to the buffer's nearest-rank quantile; then $\theta_{t+1} \leq \theta_t$ for every score sequence. **MEASURED** **0** increases in 143 and 140 steps; valve1 1.309 $\rightarrow$ **0.258**, valve2 1.126 $\rightarrow$ **0.406**.



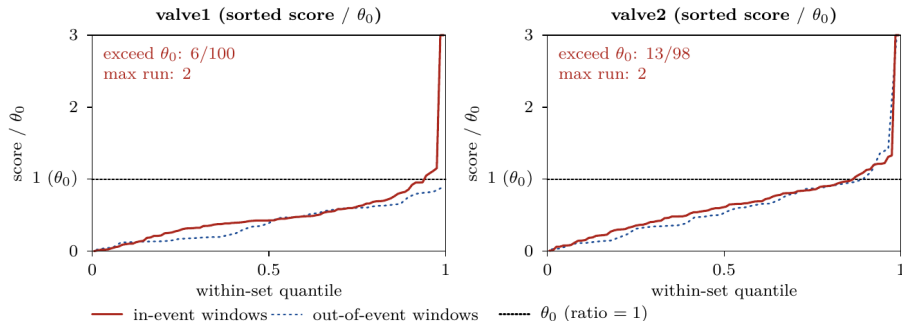
Same single event, up to 16× the alerts; our controller equals fixed

MEASURED 97 vs 6 (16.2×) and 70 vs 18 (3.9×) alert windows for the same event. Our controller: 0 deferrals, 0 recalibrations, **no harm and no evidence of benefit**. *False-alert time* = alerted samples outside the event · *recall* = events caught · *coverage* = windows decided rather than deferred.

Policy (valve1 / valve2)	Alert windows	False-alert time	Recall	Coverage
Fixed threshold	6 / 18	0 / 20	1/1	full
Rolling, normal-only	97 / 70	109 / 104	1/1	full
Rolling, admit-all	10 / 13	10 / 12	1/1	full
k-consecutive	0 / 0	0 / 0	0/1	full
Our controller	6 / 18	0 / 20	1/1	full
Sun et al.	3 / 2	0 / 4	1/1	54/143 · 57/140

The scorer, not the policy, limits detection on SKAB

MEASURED Only **6/100** (valve1) and **13/98** (valve2) in-event windows exceed θ_0 , never more than **2** in a row, so rules needing 3 miss the events. Each event spans about **70%** of replay, so recall 1/1 says little. **A policy can move alerts in time; it cannot create separation the scorer lacks.**

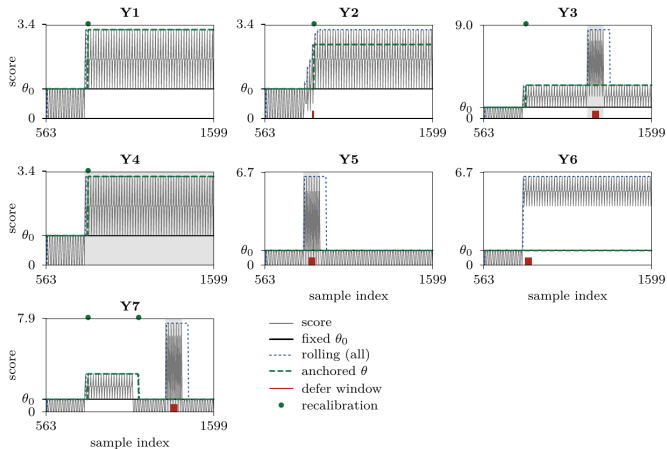


The controller: four actions, recalibration only inside a bounded band

IMPLEMENTED AnchoredRecalibrationPolicy. Its buffer holds the last 10 scores of **every** window, so what it learns from never depends on its own decisions. Nothing was tuned: 10 and 0.95 are the rolling baseline's values; the cap 4 and the stability bound 2 are declared operating assumptions.

At a window end	Action
No sustained rise	hold , or alert if $s > \theta$
Sustained rise that is stable and at most $4\theta_0$	recalibrate up to it
Sustained rise, guard fails	defer , at most 10 in a row
Still failing after 10 deferrals	escalate : alert-capable, no recalibration
Rise over	recalibrate back toward θ_0

Under stress: adapts to a step, fails on a ramp and a step-shaped fault

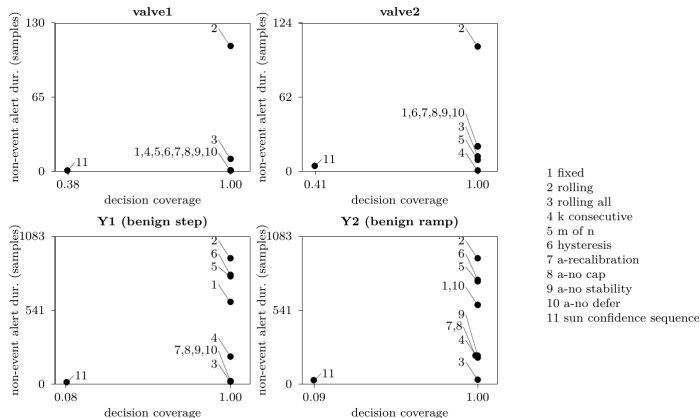


MEASURED Seven deterministic

cases:

- **Works, Y1 step:** false-alert time **16** vs fixed 597, coverage 260/260
- **Still detects, Y3 and Y7:** later faults caught, recall 1/1
- **Fails, Y2 ramp:** locks into alerts, **204** over 50 episodes
- **Fails as declared, Y4:** a fault shaped like a step is absorbed, 0.020 vs fixed 0.746

Quiet has two routes: adapt while deciding, or stop deciding



How to read it: one marker per policy; further right decides on more windows, lower means less false-alert time.

MEASURED On Y1 our controller reaches **16** while deciding on **260/260** windows.

Sun et al. are quiet because they decide on only **22/260**.

The strongest case against us: a simpler rule is quieter

These results do not establish a useful controller.

- **Admit-all is quieter on benign shifts:** false-alert time 12 vs our 16 on Y1, and 28 vs our 204 on Y2.
- **Our only edge is keeping faults visible:** on Y3 it alerts 0.29 of the fault against admit-all's 0.04, both at recall 1/1.
- **SKAB gives no evidence either way:** our rows are identical to fixed.
- **Abstention buys quiet with coverage:** Sun et al. decide on 54/143 and 57/140 windows.

What we claim instead: a proved mechanism, an honest negative-to-inconclusive development finding, and a test that can refute the direction.

Not yet shown, and the experiment designed to kill the idea

Not shown: any benefit on real benign regime changes; any held-out or generalisation result.

Limits: one event per stream, one channel (Current), dataset licence not independently verified; other/21.csv reserved and never opened.

PROPOSED Replay all policies on common traces over every other SKAB valve1 and valve2 file, admitted by a label-independent rule, plus one public family with labelled **benign** regime changes.

The direction is refuted if any one holds:

1. Where the elevation test fires, it does **not** cut false-alert time versus fixed at equal recall.
2. It **loses** an event that fixed detects.
3. Sun et al.'s abstention **matches** it at equal coverage.

A refutation becomes the reported result.